Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

R26-IT-018

Project Proposal Report

Tharaka W.A.K.N – IT 22211200

(Kothalawala L S S ,Mendis L B L , Nawarathna I K K R B)

BSc (Hons) In Information Technology Specializing in Information Technology

Department of Information Technology Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

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I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Deaf students face significant barriers in conventional e-learning environments, which predominantly rely on audio-based instruction, spoken explanations, and limited visual adaptation. In subjects such as G.C.E. Ordinary Level (O/L) Information Technology, where continuous interaction and conceptual understanding are essential, these limitations result in poor engagement, reduced motivation, and diminished academic outcomes. While prior studies have explored facial expression recognition (FER) for monitoring learner engagement in online education [1][2], and emotion detection technologies have been applied to empower deaf children in Sri Lankan contexts [5], no existing system integrates real-time emotion detection specifically for deaf O/L IT learners within an adaptive e-learning platform.

This research proposes the design and implementation of a Real-Time Emotion Detection Subsystem as a core component of a broader Accessible Emotion-Aware E-Learning Platform for deaf G.C.E. O/L IT students. The subsystem employs deep learning-based Facial Expression Recognition (FER) to classify learners' emotional states — including engagement, confusion, frustration, and neutral — in real time through webcam input. Detected emotions are communicated to the platform's adaptive content delivery module and NLP-based chatbot to trigger personalized, visually accessible learning interventions.

The system will leverage transfer learning architectures including VGG-16 and ResNet-50, trained and validated on benchmark datasets such as FER-2013 and CK+, augmented with data from voluntary deaf participants to ensure contextual accuracy [4][7]. Key performance metrics include classification accuracy, F1-score, and real-time processing latency below 200 milliseconds.

Expected outcomes include a validated emotion detection module achieving over 85% classification accuracy, seamless integration with the adaptive learning platform, and measurable improvements in engagement for deaf O/L IT learners. Socially, the system promotes inclusive education aligned with SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities). Commercially, the solution positions itself within the growing Sri Lankan EdTech market as a scalable, accessible learning tool for institutions serving deaf students.

Keywords: Facial Expression Recognition, Real-Time Emotion Detection, Deaf Education, Convolutional Neural Networks, Adaptive E-Learning

ACKNOWLEDGEMENT

I would like to acknowledge all of the people who have supported and helped me throughout the process of creating this research proposal on the Real-Time Emotion Detection Subsystem for an Accessible Emotion-Aware E-Learning Platform for Deaf Students, as this work would definitely not have been possible without their enormous support.

First and foremost, I wish to express my sincere gratitude to my supervisor, Prof. Anupama Jayalath, for her expert guidance, unwavering support, and patience throughout the entire process. Her expertise helped refine the initial research idea and provided fundamental direction at a time when I was uncertain of the path to take. Thank you for the critical feedback, insightful discussions, and continuous encouragement that were instrumental in improving the quality and depth of this work.

I would also like to extend my sincere appreciation to my research group members — Kothalawala L S S (IT22185884), Mendis L B L (IT22197764), and Nawarathna I K K R B (IT19131948) — for their collaborative efforts, stimulating discussions, and consistent encouragement throughout the proposal development process. The shared insights and constructive feedback from the team helped clarify my thoughts and significantly improved the overall quality of this work.

I wish to further thank my peers and colleagues whose encouragement and support were invaluable during this process. Their willingness to engage in thoughtful discussions and share perspectives contributed meaningfully to shaping the success of this proposal.

Finally, I wish to express my sincere appreciation to the Department of Information Technology at Sri Lanka Institute of Information Technology (SLIIT) for providing the essential academic resources, research infrastructure, and inspiring academic environment that made this work possible. This research is dedicated to the deaf student community in Sri Lanka, whose educational challenges and resilience remain the core motivation behind every aspect of this work.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ASD	Autism Spectrum Disorder
AWS	Amazon Web Services
CBAM	Convolutional Block Attention Module
CNN	Convolutional Neural Network
CK+	Extended Cohn-Kanade Dataset
CoEAI	Centre of Excellence for Artificial Intelligence
DBN	Deep Belief Network
DSR	Design Science Research
EdTech	Education Technology
EI	Engagement Index
FER	Facial Expression Recognition
FER-2013	Facial Expression Recognition Dataset 2013
FR	Functional Requirement
GCP	Google Cloud Platform
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HTTP	Hypertext Transfer Protocol
WHO	World Health Organization

1 . INTRODUCTION

1.1 Background and Context

The rapid expansion of digital education has largely been designed with hearing learners in mind, leaving deaf and hard-of-hearing students significantly underserved. Conventional elearning platforms rely heavily on audio-based explanations, spoken instructions, and video narration -modalities that are fundamentally inaccessible to deaf learners. According to the World Health Organization (WHO), approximately 466 million individuals worldwide experience disabling hearing loss, of whom 34 million are children [5]. In technically demanding subjects such as G.C.E. Ordinary Level (O/L) Information Technology, where concepts require continuous interaction and step-by-step visual explanation, this inaccessibility leads to confusion, frustration, disengagement, and ultimately poor academic outcomes. As Gupta et al. [1] establish, negative emotional states such as frustration and boredom directly divert learners' cognitive resources, causing disengagement and course abandonment in online learning environments-a risk that is considerably heightened for deaf students who lack the compensatory benefit of audio-based instructional cues.

Existing e-learning platforms such as Moodle, Coursera, and Khan Academy offer only basic accessibility features including static captions and subtitles. These solutions are non-adaptive and make no attempt to detect or respond to a learner's emotional state in real time. While significant progress has been made in Facial Expression Recognition (FER) for general online learners - Aly and Alotaibi [2] achieving 97.3% real-time classification accuracy using a hybrid ResNet-50 and 3D CNN framework, and Gupta et al. [1] reporting 92.3% accuracy using ResNet-50 on FER-2013 and CK+ datasets - none of these systems have been designed or validated for deaf learner populations. Tulane al. [3] further confirm that existing FER-based learning systems lack contextual adaptability and real-time applicability in dynamic educational settings, while Fernando et al. [5] highlight that deaf children in Sri Lanka specifically struggle with emotional comprehension and communication in digital environments, underscoring the need for a locally relevant, accessible solution.

This research proposes **a** Real-Time Emotion Detection Subsystem that captures learner facial expressions via webcam, classifies emotional states - engagement, confusion, frustration, and neutral - using deep learning architectures including VGG-16 and ResNet-50, trained on FER-2013 and CK+ datasets augmented with data from deaf participants [1][4][7]. Detected emotional states are communicated in real time to an Adaptive Content Delivery Module and an NLP-Based Chatbot, enabling the platform to dynamically adjust visual and text-based learning materials to the learner's current cognitive and emotional condition. This work directly supports SDG 4 (Quality Education) by promoting inclusive, equitable digital learning for deaf O/L IT students, and SDG 10 (Reduced Inequalities) by bridging the technological gap between hearing and deaf learners. The research falls under the CoEAI - Centre of Excellence for

Artificial Intelligence research cluster at SLIIT, targeting the Education Technology (EdTech) industry vertical at the intersection of affective computing, deep learning, and inclusive humancentered design.

1.2 Literature Review

Significant research has established FER as a reliable tool for monitoring learner engagement in online education. Dewi et al. [7] conducted a comparative analysis of FER algorithms across datasets including FER-2013, JAFFE, and CK+, confirming that deep learning approaches - particularly CNNs and DBNs - achieve recognition accuracy up to 98.82%, while highlighting persistent challenges such as lighting variation, head pose, and occlusion. Gupta et al. [1] built on this by proposing a real-time engagement detection system using InceptionV3, VGG-19, and ResNet-50, where ResNet-50 achieved 92.3% accuracy on FER-2013 and CK+ datasets, introducing an Engagement Index to classify learners as engaged or disengaged. More recently, Aly and Alotaibi [2] advanced the field with a hybrid framework combining ResNet-50, CBAM, and 3D CNN with genetic algorithm-based optimisation, achieving 97.3% real-time accuracy - demonstrating that temporal modeling of facial expressions significantly improves emotion classification in dynamic online classrooms. Hu and Gao [8] further validated FER as an engagement measurement tool, confirming a statistically significant correlation between automatically detected facial expressions and self-reported emotional engagement across six established measurement scales in synchronous online learning.

On the adaptive learning side, Tulsani et al. [3] integrated a real-time FER framework into a personalised learning system where detected states such as confusion and frustration dynamically adjusted content delivery, identifying key limitations in existing systems including insufficient real-time applicability and lack of contextual adaptability. Mohamed et al. [6] proposed the Multi-Modal Deep Attention Fusion (MMDAF) framework, fusing facial, vocal, and textual signals to personalise virtual classroom content - though its reliance on audiobased inputs makes it fundamentally inaccessible to deaf learners. Sathya and Dhatchana [10] demonstrated that CNN-based emotion classification using Grassmann feature extraction could categorise student activity states and generate institutional reports, while Alsokari and Okatan [9] showed that real-time webcam-based emotion data could be encoded into actionable instructor feedback - both systems, however, lacking any mechanism for accessible content adaptation.

The most directly relevant work for this research comes from studies targeting learners with special communication needs. Sholikah et al. [4] applied VGG-16 on FER-2013 for autism spectrum disorder (ASD) learners, achieving an F1-score of 0.91, establishing that FER can be tailored for populations with non-standard communication profiles. Fernando et al. [5] represent the only study addressing both deaf Sri Lankan children and emotion detection simultaneously, developing a CNN-based mobile application for facial emotion recognition enabling Sinhala text-to-speech communication for deaf primary students. However, their system targets general emotional communication rather than academic learning engagement and does not integrate detected emotions with adaptive e-learning content. Across all reviewed

literature, no existing system combines real-time FER with adaptive academic content delivery specifically designed and validated for deaf learners - a gap that directly motivates this research.

1.3 Research Gap

Despite considerable advancements in FER-based emotion detection for online learning, a critical and largely unaddressed gap exists at the intersection of real-time emotion detection, deaf learner accessibility, and adaptive academic content delivery. As established in the literature review, all reviewed emotion-aware e-learning systems - including those by Gupta et al. [1], Aly and Alotaibi [2], Tulsani et al. [3], and Mohamed et al. [6] - were exclusively designed and validated for hearing learners. None of these systems account for the visual-only communication needs of deaf students, nor do they adapt learning content in response to detected emotional states within a deaf-accessible interface. Furthermore, Mohamed et al.'s [6] MMDAF framework, despite its high personalisation capability, depends on audio-visual fusion inputs that are structurally inaccessible to deaf users, rendering it entirely unsuitable for the target population of this research.

Current approaches are also insufficient because they treat emotion detection and content adaptation as separate concerns, with no integrated pipeline connecting real-time affective state classification directly to accessible learning material adjustment. Tulsani et al. [3] confirm that existing FER-based learning systems lack contextual adaptability in dynamic educational settings, while Hu and Gao [8] demonstrate that FER alone cannot capture the full multidimensional nature of learner engagement without complementary adaptive mechanisms. The only study addressing deaf learners in a Sri Lankan context, Fernando et al. [5], focuses on general emotional communication for primary school children rather than curriculumaligned academic engagement for O/L IT students, leaving a clear localised gap in the literature.

The novelty of this research therefore lies in three evidence-based contributions: first, the design and validation of a real-time FER subsystem specifically calibrated for deaf O/L IT learners, using FER-2013 and CK+ benchmark datasets augmented with deaf participant data - an approach with no direct precedent in reviewed literature; second, the integration of detected emotional states into a fully accessible, visual and text-based adaptive content delivery pipeline; and third, the combination of this subsystem with an emotion-aware NLP chatbot to create a unified, end-to-end inclusive learning platform. Together, these contributions address what Dewi et al. [7] identify as a priority direction for FER research - moving beyond generic recognition systems toward context-specific, population-tailored deployments in real educational environments

Table 1: Comparison of researches with the proposed solution

Feature	Proposed Solution	Gupta et al. [1]	Aly& Alotaibi [2]	Mohamed et al. [6]	Fernando et al. [5]
Real-Time Emotion Detection	Yes	Yes	Yes	Yes	Yes

Designed for Deaf Learners	Yes	No	No	No	Partial
Visual-Only Input (No Audio)	Yes	No	No	No	Yes
Academic Curriculum Aligned (O/L IT)	Yes	No	No	No	No
Adaptive Content Delivery	Yes	No	Partial	Yes	No

1.4 Research Problem

Deaf G.C.E. O/L Information Technology students in Sri Lanka are significantly disadvantaged in existing e-learning environments due to the audio-centric design of mainstream digital education platforms. These platforms fail to detect or respond to learners' emotional states in real time, providing no mechanism to identify when a deaf student is confused, frustrated, or disengaged during an online learning session- and consequently no means to adapt the learning experience accordingly. As Fernando et al. [5] highlight, deaf students in Sri Lanka already face compounded barriers in digital communication and emotional comprehension, challenges that are directly amplified in unsupervised online learning environments where human intervention is absent.

The core research problem can therefore be stated as follows: No existing e-learning system integrates real-time facial expression-based emotion detection with adaptive, visually accessible academic content delivery specifically designed for deaf O/L Information Technology learners - resulting in persistent disengagement, poor comprehension, and inequitable learning outcomes for this population.

Current systems are insufficient on two fronts. Technically, reviewed FER systems such as those by Gupta et al. [1] and Aly and Alotaibi [2] achieve high accuracy but are neither designed nor validated for deaf learner populations, and their content adaptation mechanisms rely on audio-visual modalities inaccessible to deaf users. Socially, the absence of such a system perpetuates educational inequality, limiting the academic progression and long-term employment prospects of deaf O/L IT students - a direct violation of the inclusive education principles embedded in SDG 4 and SDG 10. This research directly addresses this dual technical and social problem through the design and implementation of a Real-Time Emotion Detection Subsystem as a core component of an accessible, emotion-aware e-learning platform tailored for this underserved population.

2 .OBJECTIVE

2.1 Main Objective

To design, implement, and validate a Real-Time Emotion Detection Subsystem that accurately classifies the emotional states of deaf G.C.E. O/L Information Technology students during online learning - including engagement, confusion, frustration, and neutral - using deep learning-based Facial Expression Recognition, and to integrate this subsystem with the broader Accessible Emotion-Aware E-Learning Platform to enable dynamic, visually accessible content adaptation and intelligent chatbot support, thereby improving learner engagement, comprehension, and academic outcomes for deaf O/L IT students in Sri Lanka.

2.2 Specific Objectives

1. **To investigate and analyze** existing Facial Expression Recognition techniques, deep learning architectures, and benchmark datasets - including FER-2013 and CK+ - to identify the most suitable approach for real-time emotion classification in a deaf learner e-learning context.

2. **To design and implement** a real-time FER pipeline using transfer learning architectures - specifically VGG-16 and ResNet-50 - capable of classifying deaf O/L IT learners' emotional states (engagement, confusion, frustration, and neutral) from live webcam input with a processing latency of under 200 milliseconds.

3. **To augment and prepare** a training dataset by combining publicly available benchmark datasets (FER-2013, CK+) with facial expression data collected from voluntary deaf participants, ensuring the model is representative of and calibrated for the target learner population.

4. **To evaluate and validate** the performance of the emotion detection model using standard classification metrics - including accuracy, precision, recall, and F1-score - targeting a minimum classification accuracy of 85% on both benchmark and participant-collected datasets.

5. **To integrate** the validated emotion detection subsystem with the Adaptive Content Delivery Module and NLP-Based Chatbot via a RESTful API, enabling real-time transmission of detected emotional states to trigger appropriate, visually accessible learning interventions.

6. **To assess** the usability and effectiveness of the integrated emotion detection subsystem through user testing with deaf O/L IT students, measuring improvements in engagement levels and comprehension outcomes compared to a non-adaptive baseline

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for projects that involve the design, construction, and evaluation of an innovative IT artifact to solve a clearly identified real-world problem [1]. DSR follows an iterative cycle of problem identification, solution design, artifact development, and evaluation -directly aligned with the objectives of this study. The research combines this with an experimental approach during the model training and validation phases, where controlled experiments are conducted to compare the performance of different deep learning architectures and measure the accuracy of emotional state classification under varying conditions.

3.2 Technical Methods and Algorithms

The emotion detection pipeline follows four sequential stages. In the face detection stage, OpenCV's Haar Cascade Classifier and the Multi-Task Cascaded CNN (MTCNN) are used to detect and localise the learner's face from the live webcam feed in real time, as applied by Gupta et al. [1] in their engagement detection system. In the preprocessing stage, detected face regions are resized to 48×48 pixels, converted to grayscale, and normalised to match the input requirements of the FER-2013 dataset. In the feature extraction and classification stage, transfer learning architectures -specifically VGG-16 **a**nd ResNet-50 **-** are fine-tuned on the combined dataset to classify emotional states into four categories: engagement, confusion, frustration, and neutral. ResNet-50 is prioritised as the primary architecture given its demonstrated 92.3% accuracy in comparable real-time learning scenarios [1] and 97.3% accuracy in hybrid configurations [2]. Finally, in the output stage, the classified emotional state is encoded as a JSON object and transmitted via RESTful API to the adaptive content module and NLP chatbot at sub-200ms latency.

3.3 Tools and Platforms

Table 2: Tools and Platforms

Tool / Framework	Purpose	Justification
Python 3.10	Core development language	Extensive ML/CV library support
TensorFlow / Keras	Model training and deployment	Industry-standard deep learning framework
OpenCV	Face detection and video capture	Lightweight, real-time CV processing
FER-2013 / CK+ Datasets	Model training and benchmarking	Widely validated in reviewed literature [1][7]
Google Colab / AWS	Model training environment	GPU-accelerated training at low cost
React.js	Front-end webcam interface	Component-based, real-time rendering

3.4 Data Collection and Ethical Considerations

Two data sources are used for model training and validation. First, publicly available benchmark datasets - FER-2013 (35,877 grayscale images across seven emotion categories) and CK+ (extended Cohn-Kanade dataset with posed and spontaneous expressions) -provide the foundational training corpus [1][7]. Second, a supplementary dataset is collected from voluntary deaf participants recruited through partner deaf schools in Sri Lanka, consisting of webcam-captured facial expression images labelled across the four target emotion categories under simulated online learning conditions.

All data collection involving human participants will adhere strictly to ethical guidelines. Informed written consent will be obtained from all participants and, where participants are minors, from their parents or guardians. Participation will be entirely voluntary with the right to withdraw at any time. All facial image data will be anonymized, securely stored with restricted access, and used solely for model training purposes. The study will comply with SLIIT institutional ethical guidelines, Sri Lanka data protection regulations, and WCAG accessibility standards. Ethical clearance will be obtained prior to any data collection activity.

3.5 Validation Strategy and Performance Metrics

Model performance will be evaluated using a stratified 80/20 train-test split, with 5-fold cross-validation applied to ensure generalisability across the combined dataset. The following metrics are used as defined success criteria:

- Classification Accuracy - minimum target of 85%, measuring overall correct emotion predictions
- Precision and Recall - evaluated per emotion class to detect class-level imbalances
- F1-Score - harmonic mean of precision and recall, used as the primary balanced metric given dataset class imbalance, as applied by Sholikah et al. [4]
- Processing Latency - real-time inference time measured per frame, target below 200ms
- Confusion Matrix - to identify misclassification patterns between emotion classes

Final system-level validation will include user acceptance testing (UAT) with deaf O/L IT students, measuring engagement improvement and usability using a structured questionnaire based on the System Usability Scale (SUS).

4 .HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Overview

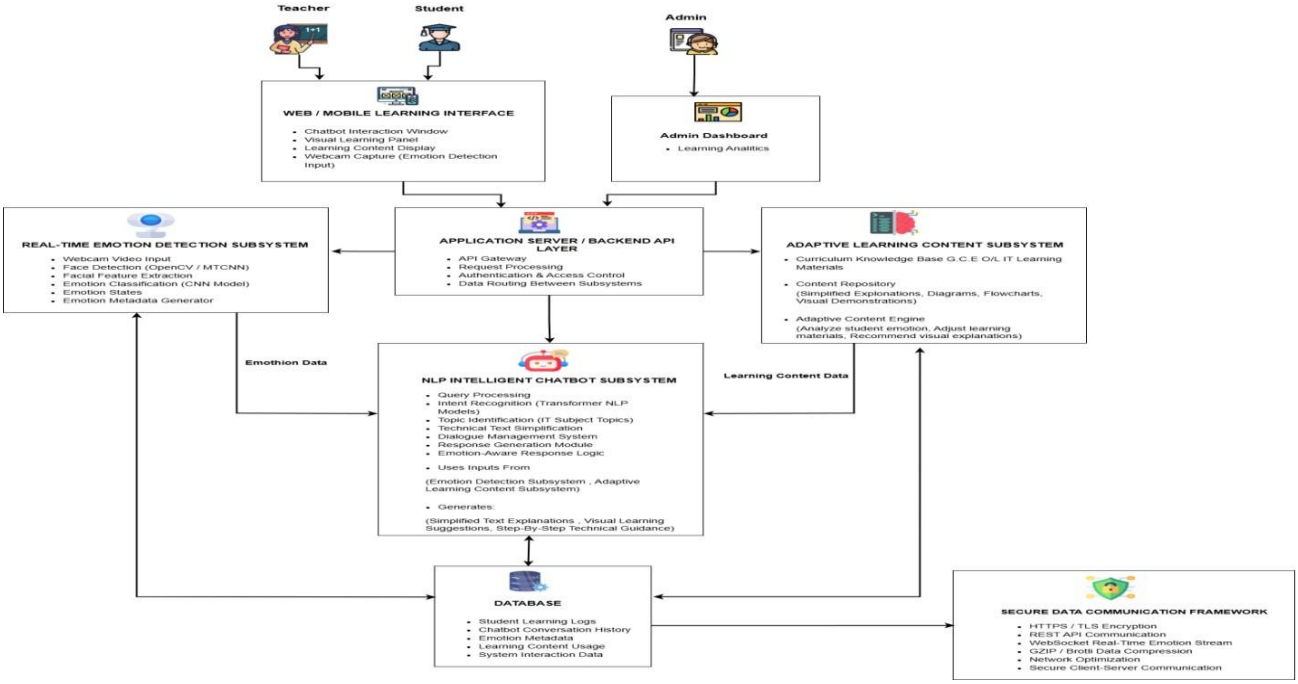
The proposed Emotion-Aware E-Learning System is designed to support deaf students studying the G.C.E. O/L Information Technology syllabus through an intelligent and adaptive learning environment. The system integrates multiple Artificial Intelligence technologies including Natural Language Processing, facial emotion recognition, and adaptive learning algorithms to provide personalized learning support.

The architecture consists of three major intelligent subsystems: the Real-Time Emotion Detection Subsystem, the Adaptive Learning Content Subsystem, and the NLP Intelligent Chatbot Subsystem. These subsystems operate together through a centralized Application Server / Backend API Layer, which manages communication, data processing, and integration between the different components of the system.

The Emotion Detection Subsystem analyzes the student's facial expressions captured through the webcam to detect emotional states such as confusion, frustration, or engagement. Based on these emotional signals, the Adaptive Learning Content Subsystem dynamically modifies the learning materials by providing simplified explanations, visual diagrams, and step-by-step demonstrations. The NLP Intelligent Chatbot Subsystem acts as an interactive tutor that answers student questions by combining the student's query, emotional state, and learning context to generate appropriate responses.

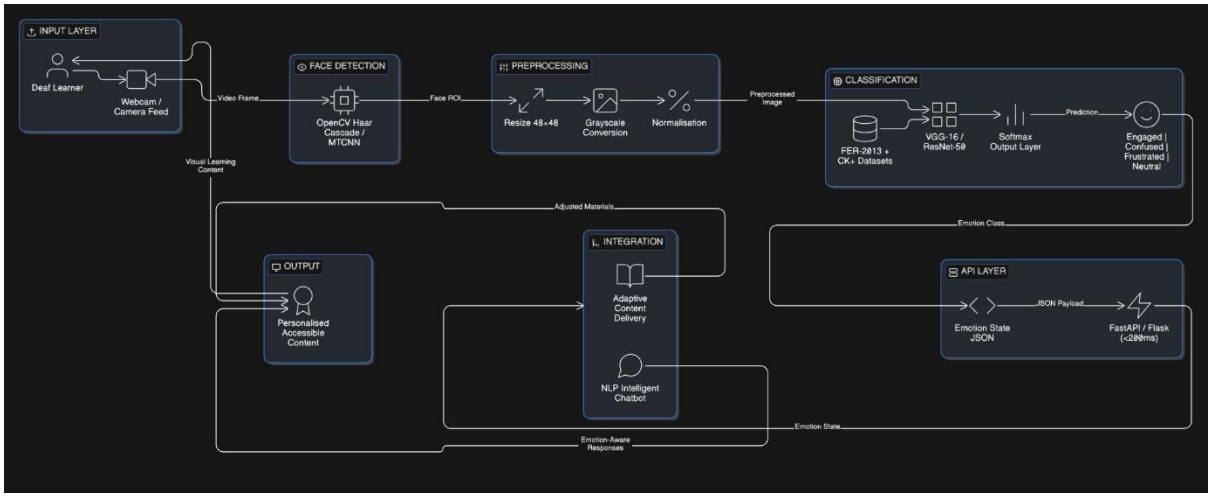
Additionally, the system includes a secure data communication framework that ensures safe and efficient transmission of information between the client interface and the backend server using encrypted communication protocols.

Figure 1: System Architectural Diagram



4.2 Component Specific System Diagram

Figure 2: Component Specific System Diagram



Input Layer - The deaf learner is captured via a standard webcam or camera feed, which continuously streams live video frames into the system. This component serves as the sole data entry point and requires no audio input, ensuring full accessibility for deaf users.

Face Detection Module- Each incoming video frame is passed through OpenCV Haar Cascade combined with MTCNN to detect and extract the learner's face region (Face ROI — Region of Interest). This module isolates the relevant facial area for further processing, filtering out background noise and irrelevant visual information.

Preprocessing Module -The extracted face ROI undergoes three sequential operations: resizing to 48×48 pixels to match dataset input specifications, grayscale conversion to reduce computational complexity, and pixel value normalisation to stabilise model training and improve inference consistency.

Classification Module - The preprocessed image is passed into the deep learning model — VGG-16 or ResNet-50 - fine-tuned on the FER-2013 and CK+ datasets. A Softmax output layer produces a probability distribution across the four target emotion classes: Engaged, Confused, Frustrated**,** and **Neutral**, with the highest probability class selected as the prediction.

API Layer - The predicted emotion class is encoded as a structured JSON payload and dispatched via FastAPI or Flask RESTful API to downstream platform components. The entire detection-to-dispatch cycle is constrained to under 200ms to maintain real-time responsiveness during active learning sessions.

Integration Layer - The JSON emotion payload is received by two parallel components: the **Adaptive Content Delivery Module**, which adjusts visual and text-based O/L IT learning materials based on the detected emotional state, and the **NLP-Based Intelligent Chatbot**, which generates emotion-aware text responses to guide and support the learner contextually.

Output Layer - The deaf learner receives personalised, visually accessible learning content dynamically adjusted to their current emotional and cognitive state, closing the feedback loop and enabling continuous, responsive learning adaptation throughout the session

4.3 Component Interactions and Data Flow

The system operates as a continuous, closed-loop pipeline during an active learning session. The deaf learner's webcam streams live video frames into the Face Detection Module, where OpenCV Haar Cascade and MTCNN isolate the facial region of interest (Face ROI) from each frame. The extracted ROI is immediately forwarded to the Preprocessing Module, which resizes the image to 48×48 pixels, applies grayscale conversion, and normalises pixel values to prepare a standardised input tensor. This preprocessed image is then passed to the Classification Module, where the fine-tuned VGG-16 or ResNet-50 model - trained on FER-2013 and CK+ datasets - applies a Softmax output layer to produce a probability distribution across the four emotion classes: Engaged, Confused, Frustrated, and Neutral. The highest-confidence class is selected as the predicted emotion state.

The prediction is immediately encoded as a structured JSON payload in the API Layer and dispatched via FastAPI or Flask RESTful API within a sub-200ms window to two downstream components in the Integration Layer: the Adaptive Content Delivery Module, which adjusts

visual and text-based learning materials accordingly, and the NLP-Based Intelligent Chatbot, which generates contextually appropriate, emotion-aware guidance for the learner. The resulting personalised, visually accessible content is returned to the learner through the Output Layer, completing the feedback loop and triggering a new cycle of emotional state monitoring.

4.4 Scalability and Security Considerations

From a scalability perspective, the modular RESTful API architecture allows the emotion detection subsystem to operate independently of platform scale. The system can be horizontally scaled by deploying multiple inference instances behind a load balancer, ensuring consistent sub-200ms latency even as concurrent user numbers increase.
Lightweight model optimisation techniques - including model quantisation and TensorFlow Lite conversion - can be applied to support deployment on lower-specification devices, broadening accessibility for schools with limited hardware infrastructure.

From a security perspective, all facial image data captured during live sessions is processed in memory and never persistently stored without explicit user consent, addressing the privacy concerns highlighted by Aly and Alotaibi [2] regarding ethical FER deployment in educational settings. All API communications between subsystem components are secured via HTTPS with token-based authentication, preventing unauthorised access to learner emotional state data. Collected participant data used for model training is fully anonymised, access-restricted, and handled in compliance with SLIIT institutional ethical guidelines and Sri Lanka data protection regulations.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The following stakeholders are directly involved in or affected by the Real-Time Emotion Detection Subsystem:

- **Deaf O/L IT Students** - primary end users whose facial expressions are captured and whose learning experience is directly shaped by the system's emotion detection output.
- **Teachers and Instructors** - secondary users who may monitor aggregated emotional engagement data to inform instructional decisions.
- **School Administrators** - institutional stakeholders responsible for platform deployment, access management, and compliance oversight.
- **System Developers** - responsible for building, maintaining, and integrating the subsystem with the broader platform.
- **Ethical Review Board / SLIIT** - oversight stakeholder ensuring data privacy, informed consent, and ethical compliance throughout data collection and system operation.

5.2 Functional Requirements

Live Webcam Video Capture The system shall capture a continuous live video stream from the learner's webcam during an active learning session.

Facial Region Detection and Extraction The system shall detect and extract the learner's facial region of interest (Face ROI) from each video frame using OpenCV Haar Cascade or MTCNN.

Face Image Preprocessing The system shall preprocess the extracted face ROI by resizing to 48×48 pixels, converting to grayscale, and normalising pixel values prior to classification.

Emotion State Classification The system shall classify the learner's emotional state into one of four categories - Engaged, Confused, Frustrated, or Neutral - using a fine-tuned VGG16 or ResNet-50 deep learning model.

Minimum Classification Accuracy Enforcement The system shall achieve a minimum emotion classification accuracy of 85% on the combined FER-2013, CK+, and participantcollected dataset.

Emotion Payload Encoding and API Transmission The system shall encode the detected emotion class as a structured JSON payload and transmit it to the Adaptive Content Delivery Module and NLP Chatbot via RESTful API.

Real-Time Processing Latency Constraint The system shall complete the full detection-todispatch cycle - from frame capture to API transmission - within 200 milliseconds per frame.

Continuous Session Monitoring The system shall operate continuously throughout an active learning session, re-evaluating the learner's emotional state at regular intervals without interruption.

Emotion Event Logging and Analytics The system shall log emotion classification events with timestamps to support session-level engagement analytics and model performance monitoring.

Audio-Free Accessibility Compliance The system shall function using only visual input, requiring no audio capture or processing, ensuring full accessibility for deaf users.

5.3 Non-Functional Requirements

Performance - The subsystem must maintain real-time processing with a maximum inference latency of 200ms per frame. Model classification accuracy must meet or exceed 85%, with an F1-score of 0.85 or above per emotion class.

Usability - The webcam-based input mechanism must require no technical configuration from the learner. The system must operate transparently in the background during a learning session without disrupting the learner's interaction with content, achieving a minimum System Usability Scale (SUS) score of 70.

Security and Privacy - Facial image data must be processed in memory only and not persistently stored without explicit informed consent. All API communications must be secured via HTTPS with token-based authentication. All participant data used for training must be anonymised and access-restricted in compliance with SLIIT ethical guidelines and Sri Lanka data protection regulations.

Reliability - The system must maintain continuous operation throughout a learning session with a minimum uptime of 99% under standard network conditions, with graceful degradation - defaulting to a neutral emotional state -in the event of camera unavailability or classification failure.

Scalability - The RESTful API architecture must support horizontal scaling to accommodate multiple concurrent users without degradation in latency or classification accuracy, targeting deployment across multiple classrooms or institutions simultaneously.

Accessibility -The system must comply with WCAG 2.1 accessibility standards and must not require any hearing-dependent input or feedback mechanism at any stage of its operation

6. COMMERSIALIZATION OF THE PRODUCT

6.1 Target Market and Customer Persona

The primary target market for the proposed Emotion-Aware E-Learning System includes educational institutions and organizations that support deaf and hard-of-hearing students. This includes special education schools, inclusive schools, and educational training centers that provide learning support for students with hearing impairments.

The system is specifically designed for students studying the **G.C.E. Ordinary Level (O/L) Information Technology syllabus,** where technical concepts often require clear explanations and visual learning materials.

A typical customer persona is a **15–16-year-old deaf student preparing for the G.C.E. O/L Information Technology examination.** These students prefer visual learning methods such as diagrams, simplified explanations, and step-by-step demonstrations instead of long text-based lessons.

Secondary users include teachers and school administrators who want to improve learning accessibility and monitor student learning progress through analytics.

6.2 Value Proposition

The proposed system provides an intelligent learning platform specifically designed to improve accessibility for deaf students. Traditional e-learning systems rely heavily on text or audio explanations, which can be difficult for deaf learners to understand.

This platform offers several key benefits:

- **Emotion-aware learning support** that detects student emotions during learning sessions.
- **Adaptive learning content** that adjusts explanations and visual materials based on the student's understanding.
- **Interactive AI chatbot tutor** that answers student questions using simplified explanations.
- **Visual learning materials** such as diagrams, flowcharts, and step-by-step demonstrations.

By combining emotion detection, adaptive learning, and NLP-based chatbot support, the system creates a more personalized and inclusive learning environment for deaf students.

6.3 Revenue Model

The system can generate revenue through several simple models suitable for an educational platform.

- **Subscription Model:**
  Educational institutions can subscribe to the platform by paying a yearly or monthly fee to allow their students to use the system.
- **Institutional Licensing:**
  Schools or educational centers can purchase a license that allows multiple students and teachers to access the system.
- **Freemium Model:**
  Basic learning support and chatbot features can be provided for free, while advanced features such as detailed learning analytics, additional visual resources, and advanced AI support can be included in a premium plan.

6.4 Cost Estimation

The development and operational costs of the system include several components as shown below.

Category	Estimated Cost (Initial Year)	Description
Development Tools	Low / Free	Use of open-source frameworks such as Python, OpenCV, TensorFlow, and NLP libraries
Cloud Hosting	$60 – $80	Basic cloud server for backend API and database
Learning Content Production	$20 - $50	Creating diagrams and visual learning materials
Database Services	$30 – $50	Hosting MongoDB database
System Maintenance	$50 - $80	System updates and bug fixes

Table 3: Cost Estimation

The use of open-source technologies helps reduce the overall cost of system development and deployment.

6.5 Pricing Strategy

The pricing strategy focuses on affordability so that educational institutions and students can easily adopt the system.

Possible pricing options include:

1. **Free Basic Plan:** Limited chatbot interaction and basic learning support.
2. **Student Plan:** Approximately $3–$5 per month for full access to chatbot tutoring and adaptive learning materials.
3. **Institutional Plan:** Approximately $200–$500 per year for schools that want to provide access to multiple students.

This pricing model ensures that the platform remains accessible while still providing sustainable funding for system maintenance and improvements.

6.5 Competitive Advantage

The proposed system offers several advantages compared with traditional e-learning platforms.

First, the system is specifically designed for **deaf and hard-of-hearing students,** focusing on visual explanations and simplified learning content.

Second, the integration of **emotion detection technology** allows the system to understand student emotional states and adjust learning materials accordingly.

Third, the system combines **emotion detection, adaptive learning, and chatbot tutoring** in a single platform, which is not commonly available in traditional e-learning systems.

These features provide a more personalized and engaging learning experience for student

6.6 Intellectual Property Considerations

The proposed system includes several components that can be protected as intellectual property.

The software architecture, chatbot algorithms, and system implementation can be protected through **software copyright protection.** Additionally, custom-developed learning materials, diagrams, and educational content can also be protected under copyright laws.

If the system demonstrates significant novelty, future research may explore applying for **patent protection** for the emotion-aware adaptive learning methodology used in the platform.

Protecting the intellectual property of the system will help ensure ownership of the developed technology and support potential commercialization opportunities in the future.

7. BUDGET AND BUDGET JUSTIFICATION

The development of the proposed Emotion-Aware E-Learning System requires several resources including hardware devices, software tools, cloud infrastructure, and human development effort. Although the project is developed within a university environment, it is important to estimate the actual cost of system development to understand the resources required if the system is implemented in a real-world environment.

The following sections present the estimated budget and justification for each cost category.

Table 2: Budget and Budget Justification

Category	**Item**	**Estimated Cost (LKR)**
Hardware	Laptop / Development Computer	250,000
Hardware	Webcam	8,000
Hardware	Internet Connectivity	6,000
Software	Open-source Development Tools (Python, OpenCV, TensorFlow, NLP Libraries)	0
Cloud Services	Cloud Hosting and Database Services	15,000
Human Effort	AI Model Development and Chatbot Implementation	200,000
Human Effort	Web Application Development and System Integration	120,000
Human Effort	System Testing and Evaluation	60,000
Total Estimated Budget		**659,000**

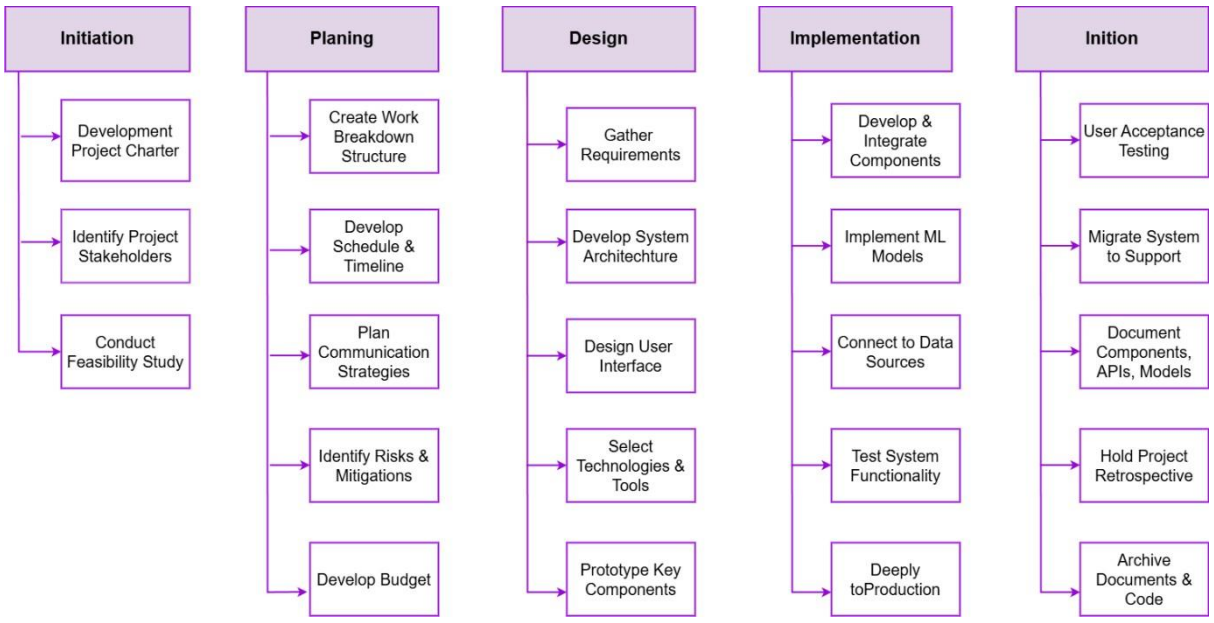
The budget mainly includes hardware resources, cloud services, and human development effort required to implement the system. A laptop is required for AI model development, software programming, and system testing. A webcam is necessary for capturing facial expressions used in the emotion detection subsystem. Cloud services are required to host the backend server, database, and learning materials so that the system can be accessed online. Human effort represents the estimated cost of software development, including AI model implementation, chatbot development, web application development, and system testing.

Open-source technologies such as Python, TensorFlow, OpenCV, and NLP libraries are used to reduce software licensing costs while still enabling advanced AI-based functionality.

8.WORK BREAKDOWN STRUCTURE

The development of the Real-Time Emotion Detection Subsystem is structured across five phases as illustrated in Figure 2. The Initiation phase covers developing the project charter, identifying stakeholders, and conducting a feasibility study to confirm technical and ethical viability. The Planning phase involves creating the work breakdown structure, developing the project schedule and timeline, planning communication strategies, identifying risks and mitigations, and finalising the project budget. The Design phase encompasses gathering system requirements, developing the system architecture, designing the user interface for the webcam integration layer, selecting tools and technologies -including TensorFlow, OpenCV, ResNet-50, and FastAPI - and prototyping key components. The Implementation phase covers developing and integrating all subsystem components, implementing and finetuning the deep learning models on FER-2013, CK+, and DAiSEE datasets, connecting data sources via RESTful API, testing system functionality, and deploying the model to production. Finally, the Initiation (closing) phase includes User Acceptance Testing with deaf O/L IT students, migrating the system to support infrastructure, documenting all components, APIs and models, conducting a project retrospective, and archiving all documents and code**.**

Figure 3: Work Breakdown Structure

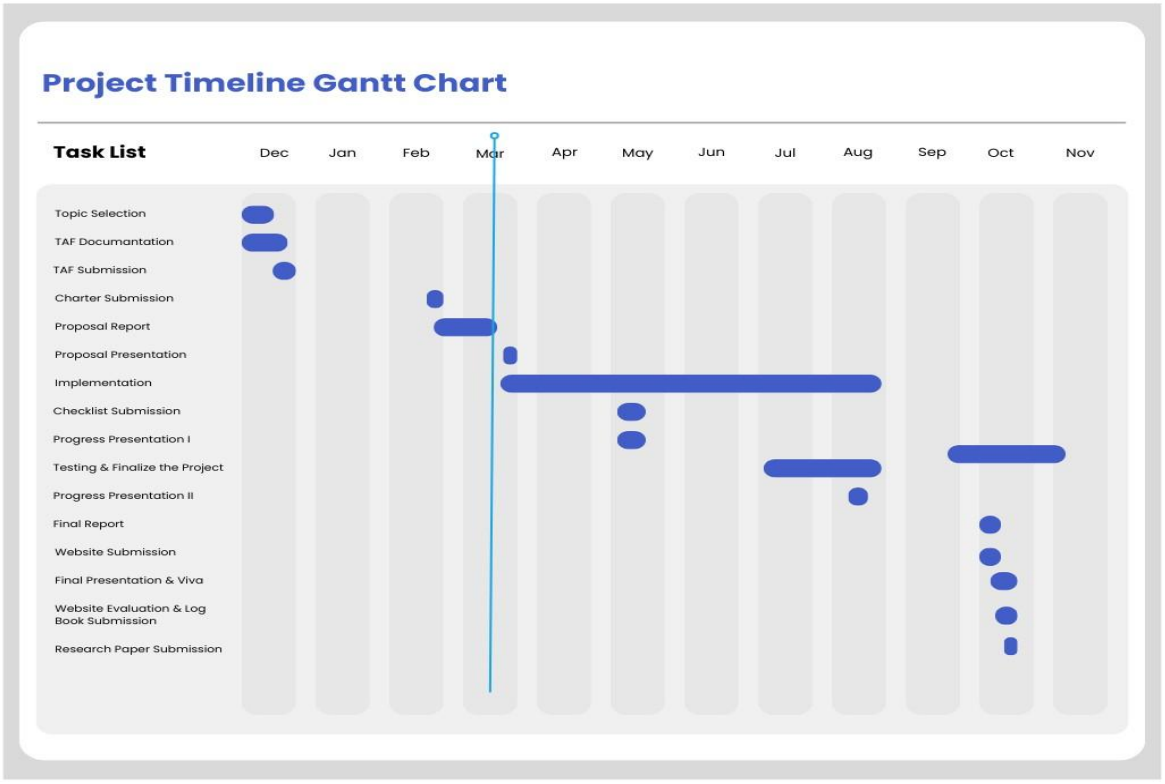


9. GANTT CHART

The project timeline spans from December 2025 to November 2026 as illustrated in Figure 3. Topic selection and TAF documentation were completed in December 2025, followed by TAF

and Charter submissions in January–February 2026. The Proposal Report and Proposal Presentation were completed in March 2026, marking the end of the planning phase. The Implementation phase runs from April through August 2026, covering model development, dataset preparation, system integration, and API deployment, with a Checklist Submission and Progress Presentation I scheduled in May 2026. Testing and project finalisation spans July to August 2026, with Progress Presentation II in August 2026. The final phase from September to November 2026 covers the Final Report, Website Submission, Final Presentation and Viva, Website Evaluation and Log Book Submission, and Research Paper Submission — completing the full project lifecycle within a twelve-month timeline

Figure 4 : Grantt Chart



REFERENCES

[1] Gupta et al. (2023) - *Facial emotion recognition based real-time learner engagement detection* - Multimedia Tools & Applications https://doi.org/10.1007/s11042-022-13558-9

[2] Aly & Alotaibi (2025) - *Deep learning framework for real-time emotion detection in online learning* -Scientific Reportshttps://doi.org/10.1038/s41598-025-26381-7

[3] Tulsani et al. (2025) - *Real-Time Facial Emotion Recognition for AI-Enhanced Personalized Learning* - JISEMhttps://www.jisem-journal.com

[4] Sholikah et al. (2024) - *Real-time FER to Enhance Emotional Intelligence in Autism* — Procedia CShttps://doi.org/10.1016/j.procs.2024.02.169

[5] Fernando et al. (2023) -*Empowering Deaf Children with Sinhala Sign Language, Emotion Detection* - IEEE ICAChttps://doi.org/10.1109/ICAC60630.2023.10417483

[6] Mohamed et al. (2025) - *Emotion-Aware Course Personalization via Multi-Modal Fusion* - IEEE ICCRhttps://doi.org/10.1109/ICCR67387.2025.11291875

[7] Dewi et al. (2024) - *Real-Time FER: Advances, Challenges, Future Directions* -Vietnam J. Computer Sciencehttps://doi.org/10.1142/S219688882330003X

[8] Hu & Gao (2025) - *FER reveals students' engagement in online class* -PLoS ONEhttps://doi.org/10.1371/journal.pone.0334232

[9] Zhang et al. (2016) -*Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks* - IEEE Signal Processing Letters https://doi.org/10.1109/LSP.2016.2603342

[10] Ikram et al. (2023) - *Recognition of Student Engagement State Using Deep and Efficient Transfer Learning - VGG16* - Applied Sciences https://doi.org/10.3390/app13158637

[11] Mahmood et al. (2024) - *Measuring Student Engagement through Behavioral and Emotional Features Using Deep-Learning Models* - MDPI Algorithms https://doi.org/10.3390/a17100458

[13] Alsokari & Okatan (2023) -*Real-Time Online Education Student Lecture Emotion Detection* - EJEAS https://doi.org/10.17932/EJEAS.2021.024/ejeas_v03i1001